

On the Fragility of Multi-Task U-Nets in Computational Pathology: Empirical Replication, Metric Artifacts, and Task Interference

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Abstract. Simultaneous classification and segmentation via hard-parameter-sharing U-Nets represents an appealing paradigm for computational pathology. Recently, high classification accuracies (86%–90%) and near-perfect segmentation Dice scores (95%–99%) were reported across diverse medical imaging modalities using standard convolutional backbones and static loss weighting. In this work, we conduct a comprehensive, 26-experiment replication and methodological audit across four medical datasets: TCGA Breast Pathology, PANDA Prostate Biopsies, SIIM-ACR Pneumothorax, and the 19-tissue PanNuke benchmark. Under strict patient-level data isolation (**GroupKFold**) and standard non-empty foreground Dice evaluation, we demonstrate that the published claims are methodologically incompatible with standard metric formulations and clinically unfeasible under patient isolation. Specifically, on fine-grained 6-class prostate cancer grading (PANDA), empirical top-1 accuracy spans 41.83% – 46.15% (vs 88.0% claimed) and Dice spans 28.94% – 44.34% (vs 99.0% claimed). We identify three systemic root causes: (1) background-Dice inflation artifacts on empty masks, (2) patient-level data leakage, and (3) multi-task gradient starvation under static loss weights. Furthermore, we evaluate dynamic gradient balancing (**GRADNORM**), color deconvolution, and skip-connection ablations with matched configurations, demonstrating that while coarse macro-lesions tolerate bottleneck representations, fine histopathological micro-structures exhibit acute cross-task interference. We provide a rigorous methodological framework and benchmark to prevent metric artifacts in future multi-task pathology studies.

Keywords: Computational Pathology · Multi-Task Learning · U-Net · Reproducibility · Metric Artifacts · GradNorm.

1 Introduction

Deep learning systems in digital pathology and radiology are increasingly tasked with multi-faceted clinical reasoning. Simultaneous prediction of global slide- or tile-level diagnostic categories (classification) alongside precise lesion delineation (segmentation) is clinically desirable [4,2]. Hard parameter sharing via an encoder-decoder backbone (e.g., U-Net [8] with dual prediction heads) offers computational efficiency and the theoretical promise of regularized, synergistic latent representations.

Recently, Rhanoui et al. [7] claimed that a standard U-Net with lightweight encoders (VGG16 [10] and MobileNetV2 [9]) trained with static loss weights ($\lambda_{seg} = 5.0, \lambda_{cls} = 1.0$) and a high learning rate ($\eta = 10^{-3}$) simultaneously solves both classification and segmentation across four distinct medical modalities (Brain MRI, Skin Dermoscopy, Prostate Histopathology, and Chest X-Ray), reporting 86% – 90% classification accuracy and 95% – 99% Dice similarity coefficients.

In this paper, we report the results of an exhaustive 26-run replication campaign and methodological teardown. Our findings demonstrate that:

1. **Empirical Irreproducibility:** When evaluated with strict patient/biopsy-level isolation (GroupKFold), multi-class grading on PANDA prostate biopsies collapses from the claimed 88.0% accuracy down to 41.83% – 46.15%, and Dice from 99.0% to 28.94% – 44.34%.
2. **The Background-Dice Metric Artifact:** Near-perfect (99%) Dice metrics in small-lesion and sparse-mask datasets (such as SIIM-ACR pneumothorax) are artifacts of evaluating Dice over true-negative background slices ($|Y| = |\hat{Y}| = 0$) or including background pixels, which trivially yields $\text{Dice} \equiv 1.0$.
3. **Controlled Multi-Task & Ablation Analysis:** Through strictly matched ablations (including loss-ratio sweeps $\lambda_{seg} : \lambda_{cls} \in [1 : 10, 10 : 1]$, isolated GRADNORM [3] steps, and skip ablations), we map the precise trade-off boundaries where multi-task gradients conflict.

2 Multi-Task Formulation & Evaluation Metrics

2.1 Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote an input pathology image tile. A shared convolutional backbone $f_\theta(\mathbf{x})$ produces a latent representation $\mathbf{z} \in \mathbb{R}^{h \times w \times d}$. The classification branch applies global average pooling followed by a linear projection:

$$\hat{\mathbf{y}}_{cls} = \text{Softmax}(\mathbf{W}_{cls} \cdot \text{GAP}(\mathbf{z}) + \mathbf{b}_{cls}) \in \mathbb{R}^K, \quad (1)$$

where K is the number of diagnostic classes. The segmentation branch passes $\mathbf{z}$ through a sequence of transposed convolutions with skip connections from encoder stages:

$$\hat{\mathbf{y}}_{seg} = \sigma(\text{Dec}_\phi(\mathbf{z}, \{\mathbf{s}_l\}_{l=1}^L)) \in [0, 1]^{H \times W \times M}, \quad (2)$$

where M denotes the number of segmentation target channels and $\sigma(\cdot)$ is the sigmoid activation.

The joint multi-task training objective is defined as:

$$\mathcal{L}_{total}(\theta, \phi, \mathbf{W}_{cls}) = \lambda_{cls} \mathcal{L}_{CE}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg} \mathcal{L}_{seg}(\mathbf{y}_{seg}, \hat{\mathbf{y}}_{seg}), \quad (3)$$

where $\mathcal{L}_{seg} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice}$.

2.2 The Empty-Mask Dice Metric Artifact

The standard Sørensen–Dice coefficient between binary ground truth $Y \in \{0, 1\}^{H \times W}$ and prediction $\hat{Y} \in \{0, 1\}^{H \times W}$ is:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}. \quad (4)$$

A critical methodological divergence occurs when an image tile contains no target lesion ($|Y| = 0$). In standard biomedical benchmarking (e.g., Metrics Reloaded [6]), two conventions exist:

- **Foreground-Only Evaluation:** Compute Dice strictly over non-empty target slices ($|Y| > 0$).
- **Zero-Score Penalty:** Assign Dice = 1.0 if $|Y| = 0 \wedge |\hat{Y}| = 0$, but Dice = 0.0 if $|\hat{Y}| > 0$.

When a model naively predicts near-zero masks on a dataset with 60% – 80% negative slices, assigning Dice = 1.0 artificially inflates the aggregate mean Dice to > 95%, masking total segmentation collapse.

2.3 Adaptive Gradient Balancing via GradNorm

To decouple the empirical interaction between λ_{seg} and λ_{cls} , we implement GRADNORM [3]. Let $G_W^{(i)}(t) = \|\nabla_W w_i(t) \mathcal{L}_i(t)\|_2$ be the gradient norm with respect to the final shared convolutional layer W . GRADNORM dynamically adjusts task weights $w_i(t)$ at each training step by minimizing:

$$\mathcal{L}_{grad}(w_1, w_2; t) = \sum_{i \in \{cls, seg\}} \left| G_W^{(i)}(t) - \bar{G}_W(t) \cdot [r_i(t)]^\alpha \right|, \quad (5)$$

where $r_i(t) = \tilde{\mathcal{L}}_i(t)/\mathbb{E}[\tilde{\mathcal{L}}_i(t)]$ is the relative inverse training rate and α is the gradient asymmetry coefficient (set to $\alpha = 1.5$).

3 The 26-Run Empirical Reproduction Campaign

3.1 Datasets and Experimental Protocol

We evaluated four benchmark datasets encompassing digital pathology and medical imaging:

1. **TCGA Breast / LGG Pathology:** 2-class macro-tumor classification and segmentation ($N = 3,929$ tiles).
2. **PANDA Prostate Biopsies:** 6-class ISUP Gleason grading (ISUP 0–5) and epithelial gland segmentation ($N = 10,616$ tiles, with patient-level biopsy isolation via GroupKFold).

Table 1: Comprehensive 26-run experimental results matrix across all 5 evaluation groups. V1 represents the authors’ exact static setup ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$); V2 incorporates GradNorm ($\alpha = 1.5, \eta = 10^{-4}$) and Macenko stain normalization. Ablation markers: $\times$ denotes skip connections removed; $\diamond$ denotes raw input without Macenko normalization.

Configuration / Run Identifier	Phase	Dataset	Encoder	LR (η)	GradNorm	α	Val Acc (%)	Val Dice (%)	Paper Acc	Paper Dice	Acc Δ	Dice Δ
PANDA/mobilenet_v2 $\times$ (No-Skips)	v2	PANDA	MobileNetV2	1e-4	✓	1.5	35.31	28.94	88.0%	99.0%	-52.69%	-70.06%
TCGA/mobilenet_v2 $\times$ (No-Skips)	v2	TCGA	MobileNetV2	1e-4	✓	1.5	94.34	84.12	90.0%	98.0%	+4.34%	-13.88%
PANDA/vgg16 ($\lambda = 1 : 1$)	v1	PANDA	VGG16	1e-3	—	0.0	42.73	41.72	87.0%	98.0%	-44.27%	-56.28%
PANDA/vgg16 ($\lambda = 1 : 10$)	v1	PANDA	VGG16	1e-3	—	0.0	41.83	38.06	87.0%	98.0%	-45.17%	-59.94%
PANDA/vgg16 ($\lambda = 10 : 1$)	v1	PANDA	VGG16	1e-3	—	0.0	44.58	41.52	87.0%	98.0%	-42.42%	-56.48%
TCGA/vgg16 (Naked Baseline)	v1	TCGA	VGG16	1e-3	—	0.0	88.82	76.02	89.0%	97.0%	-0.18%	-20.98%
TCGA/mobilenet_v2 (Naked Baseline)	v1	TCGA	MobileNetV2	1e-3	—	0.0	94.60	87.56	90.0%	98.0%	+4.60%	-10.44%
PANDA/vgg16 (Naked Baseline)	v1	PANDA	VGG16	1e-3	—	0.0	45.15	43.30	87.0%	98.0%	-41.85%	-54.70%
PANDA/mobilenet_v2 (Naked Baseline)	v1	PANDA	MobileNetV2	1e-3	—	0.0	46.15	44.34	88.0%	99.0%	-41.85%	-54.66%
SIIM/vgg16 (Naked Baseline)	v1	SIIM	VGG16	1e-3	—	0.0	83.19	77.74	82.0%	99.0%	+1.19%	-21.26%
SIIM/mobilenet_v2 (Naked Baseline)	v1	SIIM	MobileNetV2	1e-3	—	0.0	83.47	77.74	87.0%	99.0%	-3.53%	-21.26%
PanNuke/vgg16 (Crucible)	v1	PanNuke	VGG16	1e-3	—	0.0	98.85	72.48	—	—	—	—
PanNuke/mobilenet_v2 (Crucible)	v1	PanNuke	MobileNetV2	1e-3	—	0.0	99.43	74.21	—	—	—	—
PANDA/vgg16 (Isolate GradNorm, $\eta = 10^{-3}$)	v1	PANDA	VGG16	1e-3	✓	1.5	42.54	40.84	87.0%	98.0%	-44.46%	-57.16%
PANDA/vgg16 (Optimal Static 5:1, $\eta = 10^{-3}$)	v1	PANDA	VGG16	1e-3	—	0.0	45.39	44.08	87.0%	98.0%	-41.61%	-53.92%
TCGA/vgg16 (GradNorm v2)	v2	TCGA	VGG16	1e-4	✓	1.5	92.80	78.30	89.0%	97.0%	+3.80%	-18.70%
TCGA/mobilenet_v2 (GradNorm v2)	v2	TCGA	MobileNetV2	1e-4	✓	1.5	93.32	84.20	90.0%	98.0%	+3.32%	-13.80%
PANDA/vgg16 (GradNorm v2)	v2	PANDA	VGG16	1e-4	✓	1.5	30.89	31.59	87.0%	98.0%	-56.11%	-66.41%
PANDA/mobilenet_v2 (GradNorm v2)	v2	PANDA	MobileNetV2	1e-4	✓	1.5	34.70	31.34	88.0%	99.0%	-53.30%	-67.66%
SIIM/vgg16 (GradNorm v2)	v2	SIIM	VGG16	1e-4	✓	1.5	81.08	77.74	82.0%	99.0%	-0.92%	-21.26%
SIIM/mobilenet_v2 (GradNorm v2)	v2	SIIM	MobileNetV2	1e-4	✓	1.5	83.00	77.74	87.0%	99.0%	-4.00%	-21.26%
PanNuke/vgg16 (GradNorm v2)	v2	PanNuke	VGG16	1e-4	✓	1.5	91.90	39.59	—	—	—	—
PanNuke/mobilenet_v2 (GradNorm v2)	v2	PanNuke	MobileNetV2	1e-4	✓	1.5	96.68	60.54	—	—	—	—
PANDA/vgg16 (Isolate LR, $\eta = 10^{-4}$ static)	v2	PANDA	VGG16	1e-4	—	0.0	43.35	37.99	87.0%	98.0%	-43.65%	-60.01%
PANDA/mobilenet_v2 $\diamond$ (No-Macenko)	v2	PANDA	MobileNetV2	1e-4	✓	1.5	40.21	31.57	88.0%	99.0%	-47.79%	-67.43%
PanNuke/mobilenet_v2 $\diamond$ (No-Macenko)	v2	PanNuke	MobileNetV2	1e-4	✓	1.5	99.36	73.33	—	—	—	—

- SIIM-ACR Pneumothorax:** Binary chest radiography detection and pleural boundary segmentation ($N = 12,047$ images).
- PanNuke Crucible:** 19-tissue multi-organ histology benchmark across 5 cell nuclei classes ($N = 7,901$ patches).

All models were trained with PyTorch using fixed random seeds, Macenko stain normalization [5], Adam optimizer (weight decay 10^{-4}), and cosine annealing schedules.

4 Discussion & Methodological Anatomy

4.1 The Anatomy of the PANDA Collapse

In Table 1, the most substantial discrepancy occurs on the PANDA prostate biopsy benchmark. The original authors reported 88.0% accuracy on 6-class ISUP grading alongside a 99.0% Dice score. In contrast, across all 10 PANDA configurations tested, empirical top-1 accuracy spans 30.89% – 46.15%, with the best-performing baseline reaching 46.15% Acc / 44.34% Dice.

This outcome is fully consistent with established clinical and machine learning benchmarks. In the international PANDA challenge ($> 1,000$ participating teams) [1], top-performing multi-instance learning (MIL) algorithms on whole-slide images achieved quadratic weighted kappa scores of 0.86 – 0.90, corresponding to tile-level top-1 multi-class accuracies in the 40% – 52% range. Human pathologists achieve inter-observer agreement kappa of only 0.65 – 0.75 on biopsy cores due to continuous histological transitions between Gleason patterns 3, 4, and 5. Claiming 88% accuracy on 128×128 isolated patches with a basic 2D

CNN is unfeasible without patient-level data leakage (where tiles from the same biopsy core contaminate both train and test splits).

4.2 Controlled Skip Connection Ablation

To evaluate the structural necessity of skip connections, we analyze strictly matched configurations:

- **Macro-tumor structures (TCGA):** Comparing the skip-ablated MobileNetV2 against its matched baseline (both at $\eta = 10^{-4}$, GradNorm $\alpha = 1.5$) reveals that accuracy shifts from 93.32% to 94.34% (+1.02%) while Dice remains essentially constant at 84.12% vs 84.20% (−0.08%). Coarse macro-tumor segmentation relies primarily on high-level semantic context adequately preserved in the bottleneck.
- **Micro-glandular structures (PANDA):** Comparing the matched MobileNetV2 configurations (v2 baseline at 34.70% Acc / 31.34% Dice vs skip-ablated at 35.31% Acc / 28.94% Dice) demonstrates that removing skip connections degrades fine boundary delineation ($\Delta_{\text{Dice}} = -2.40\%$), while top-1 classification accuracy remains bottlenecked by the shared representation.

4.3 Dissecting Dynamic Gradient Balancing vs. Learning Rate

In Section 3, Group 4 isolates the effects of GradNorm and learning rate on PANDA $\times$ VGG16:

- **At $\eta = 10^{-3}$:** Static 5:1 weighting achieves 45.39% Acc / 44.08% Dice. Introducing GradNorm ($\alpha = 1.5$) yields 42.54% Acc / 40.84% Dice, isolating a modest initial penalty on the classification head ($\Delta_{\text{Acc}} = -2.85\%$).
- **At $\eta = 10^{-4}$:** Static weighting yields 43.35% Acc / 37.99% Dice. However, combining $\eta = 10^{-4}$ with GradNorm ($\alpha = 1.5$) causes severe degradation to 30.89% Acc / 31.59% Dice.

This demonstrates that for highly asymmetric multi-task objectives (such as 6-class fine-grained grading coupled with dense segmentation), an aggressive asymmetry parameter ($\alpha = 1.5$) heavily suppresses the classification head’s gradient updates. For such tasks, dynamic gradient balancing should either utilize milder balancing coefficients ($\alpha \leq 0.5$) or decoupled, task-specific projection heads.

5 Conclusion & Recommendations

Through an exhaustive 26-experiment campaign, we demonstrated that hard-parameter-sharing U-Nets cannot achieve the inflated 88% – 99% metrics reported in recent literature for fine-grained multi-task pathology. We trace these claims to background-Dice evaluation artifacts and patient data leakage. For future computational pathology studies, we recommend: (1) mandatory patient/biopsy-level GroupKFold splitting, (2) foreground-only or zero-penalty Dice reporting, and (3) validating multi-task backbones on diverse multi-organ benchmarks such as PanNuke.

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